task_4 .py - C:/Users/josth/AppData/Local/Programs/Python/Python313/task_4 .py (3.13.17)

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```python
import math

# TASK 4 - Write Correct Expressions

# Constants
BASE_RATE = 30.0
FAT_STEP = 0.1
SNF_STEP = 0.1
FAT_RATE = 0.40
SNF_RATE = 0.30

# Milk details
fat = 4.2
snf = 3.1
quantity = 100

# Calculate number of steps
fat_steps = round((fat - 3.5) / FAT_STEP)
snf_steps = round((snf - 3.0) / SNF_STEP)

# Calculate rate per litre
rate_per_litre = (
    BASE_RATE
    + fat_steps * FAT_RATE
    + snf_steps * SNF_RATE
)

# Calculate total amount
amount = quantity * rate_per_litre

print("Fat:", fat)
print("SNF:", snf)
print("Fat Steps:", fat_steps)
print("SNF Steps:", snf_steps)
print("Rate per Litre: Rs.", format(rate_per_litre, ".2f"))
print("Total Amount: Rs.", format(amount, ".2f"))

# Unexpected floating-point results
print("\nUNEXPECTED RESULTS")
print("(4.2 - 3.5) / 0.1 =", (4.2 - 3.5) / 0.1)
print("0.1 + 0.2 =", 0.1 + 0.2)
print("0.1 + 0.2 == 0.3:", 0.1 + 0.2 == 0.3)
```

```python
# Operator precedence and associativity
print("\nOPERATOR PRECEDENCE")
print("2 ** 3 ** 2 =", 2 ** 3 ** 2)
print("-2 ** 2 =", -2 ** 2)
print("(-2) ** 2 =", (-2) ** 2)

# Comparison chaining and Boolean operators
print("\nCOMPARISONS")
print("Fat valid:", 2.0 <= fat <= 7.0)
print("SNF valid:", 3.0 <= snf <= 9.0)
print("Both valid:",
      2.0 <= fat <= 7.0 and 3.0 <= snf <= 9.0)

# Compare round(), int(), and math.floor()
print("\nROUND, INT AND FLOOR")

value = (4.2 - 3.5) / 0.1
print("Value:", value)
print("round:", round(value))
print("int:", int(value))
print("floor:", math.floor(value))

negative_value = -2.7
print("\nNegative value:", negative_value)
print("round:", round(negative_value))
print("int:", int(negative_value))
print("floor:", math.floor(negative_value))

# Compare step calculation methods
print("\nSTEP CALCULATION COMPARISON")

for fat_value in (4.2, 3.1):
    value = (fat_value - 3.5) / 0.1

    print("\nFat:", fat_value)
    print("Raw steps:", value)
    print("round steps:", round(value))
    print("int steps:", int(value))
    print("floor steps:", math.floor(value))
```

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    print("floor steps:", math.floor(value))

    print("Rate using round:",
          format(BASE_RATE + round(value) * FAT_RATE, ".2f"))
    print("Rate using int:",
          format(BASE_RATE + int(value) * FAT_RATE, ".2f"))
    print("Rate using floor:",
          format(BASE_RATE + math.floor(value) * FAT_RATE, ".2f"))

# Complexity analysis
print("\nCOMPLEXITY ANALYSIS")
print("Time complexity: O(1)")
print("Space complexity: O(1)")
```